28/9/26

PRACTICAL - 7

Aim:

To implement and stimulate the sliding window protocol using C programming to demonstrate frame translation, acknowledgement (ACK), negative acknoculedgments (NACK), and retransmission handling using a circular queue.

code:

```
#include<stdio.h>
#include<string.h>
#define MAX 100
struct Frame{
    int frameNo;
    char data;
};
int main(){
    struct Frame frames[MAX];
    char message[MAX];
    int totalFrames, windowSize;
    int base = 0;
    int expectedFrame = 1;
    int loseFrame;
    int lossSimulated = 0;
    int i, end;
    int needRetransmission;
    printf("Enter the message:");
    fgets(message, sizeof(message), stdin);
    message[strcspn(message,"\n")]='\0';
    totalFrames = strlen(message);
```

```
if (totalFrames == 0){
    printf ("Message cannot be empty.\n");
    return 1;
}
printf ("Enter window size: ");
if (scanf ("%d", &windowSize) != 1 ||
    windowSize <= 0 || windowSize > MAX){
    printf ("Invalid window size.\n");
    return 1;
}
if (windowSize > totalFrames){
  windowSize = totalFrames;
}
printf ("Enter frame number to lose (0 for none):");
if (scanf ("%d", &lostFrame) != 1 ||
    lostFrame < 0 || lostFrame > totalFrames){
    printf("Invalid frame number.\n");
    return 1;
}
for (i= 0; i< totalFrames; i++){
    frames [i]. frameNo = i+1;
    frames[i]. data = message [i];
}
printf ("\n SLIDING WINDOW PROTOCOL \n");
printf ("Total frame: %d\n", totalFrames);
printf ("WindowSize : %d\n",windowSize);
while (base < totalFrames){
    end = base + windowSize;
    if ( end > totalFrames) {
        end = totalFrames;
    }
    need retransmission = 0;
```

```
printf ("SENDER WINDOW:");
for (i= base ; i< end ; i++){
    printf ("[%.d : %.c]", frames[i] . frameNo,
        frames[i]. data);
}
print("\n SENDER TRANSMITS FRAMES \n");
for (i= base ; i< end ; i++){
    printf(" Sender -> Frame %.d , Data : %.c \n",
        frames[i] . frameNo , frames[i] . data);
}
printf ("\n RECEIVER PROCESSES FRAMES \n");
for (i= base ; i< end ; i++){
    int frameNo = frames [i]. frameNo;
    if (frameNo == lostFrame && ! lossSimulated){
        printf ("Frame %.d was lost during transmission.
                                        \n";
            frameNo);
        lossSimulated = 1;
        needRetransmission= 1;
        printf ("Receiver -> NACk %.d \n", expected
                Frame);
        continue;
    }
    printf ("Receiver <- Frame %d, Data : %.c \n",
            frameNo , frames[i].data);
    if (frameNo == expected Frame){
        printf ("Receiver -> ACk %.d \n",
                frameNo);
        expectedFrame ++;
}
```

```
        else{
            printf("Receiver discards out-of-order
                frame %d.\n", frameNo);
            printf("Receiver -> NACK %d\n",
                expectedFrame);
            needRetransmission = 1;
        }
    }
    if (needRetransmission){
        printf("\n GO-BACK-N RETRANSMISSION\n");
        for (i = expectedFrame - 1; i < end; i++){
            printf("Sender retransmits Frame %d,
                Data: %c\n", frames[i].
                frameNo, frames[i].data);
        }
        printf("\n RECEIVER PROCESSES
                RETRANSMITTED FRAMES \n");
        for (i = expectedFrame - 1; i < end; i++){
            int frameNo = frames[i].frameNo;
            printf("Receiver <- Frame %d,
                Data: %c\n", frameNo,
                frame[i].data);
            if (frameNo == expectedFrame){
                printf("Receiver -> ACK %d\n",
                    frameNo);
                expectedFrame++;
            }else{
                printf("Receiver -> NACK %d\n",
                    expectedFrame);
            }
        }
    }
}
```

```
        base = end;
        if (base < totalFrames){
            printf("\n Window slides forward to frame
                %d \n", frames[base].frameNo);
        }
    }
    printf("All frames transmitted and acknowledged
                \n");

    printf("Original message : %s\n", message);

    return 0;

}
```

Output:

Enter the message : Hello

Enter window size : 3

Enter frame number to lose (0 for none) : 2

SLIDING WINDOW PROTOCOL

Total frames: 5

Window size : 3

SENDER WINDOW: [1:H] [2:e] [3:l]

SENDER TRANSMITS FRAMES

Sender -> Frame 1, Data: H

Sender -> Frame 2, Data: e

Sender -> Frame 3, Data: l

RECEIVER PROCESSES FRAMES:

Receiver <- Frame 1, Data: H

Receiver -> ACK 1

Frame 2 was lost during transmission.

Receiver -> NACK 2

Receiver <- Frame 3, Data : l

Receiver discards out-of-order frame 3.

Receiver -> NACK 2

GO-BACK-N RETRANSMISSION

Sender retransmits Frame 2, Data : e

Sender retransmits Frame 3, Data : l

RECEIVER PROCESSES RETRANSMITTED FRAMES

Receiver <- Frame 2, Data : e

Receiver -> ACK 2

Receiver <- Frame 3, Data : l

Receiver -> ACK 3

Window slides forward to frame 4.

SENDER WINDOW: [4: l] [5: o]

SENDER TRANSMITS FRAMES

Sender -> Frame 4, Data : l

Sender -> Frame 5, Data : o

RECEIVER PROCESSES FRAMES

Receiver <- Frame 4, Data : l

Receiver -> ACK 4

Receiver <- Frame 5, Data : o

Receiver -> ACK 5

All frames transmitted and acknowledged.

Original message : Hello

Result

The flow control using sliding window protocol was learned and executed successfully.